

1. Tasks (Taxonomy)



Office & Productivity

Spreadsheets, Documents, Presentations, Emails & Messaging



Web & Information

Public Reference, Shopping & Commerce, Travel & Local, Academic Lookup, Web Archiving



System & Software Operations

Application & Env Config, OS & File Operations, Software Development



Scientific & Engineering

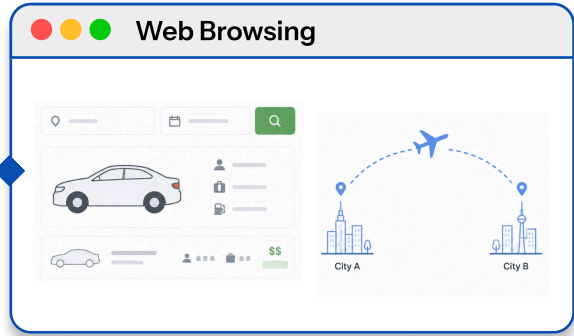
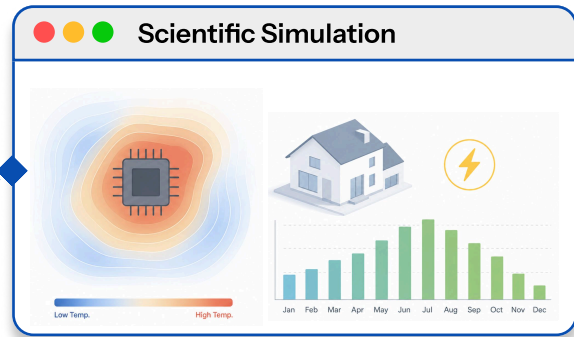
Engineering Simulation, Medical Imaging, Bioimage Analysis



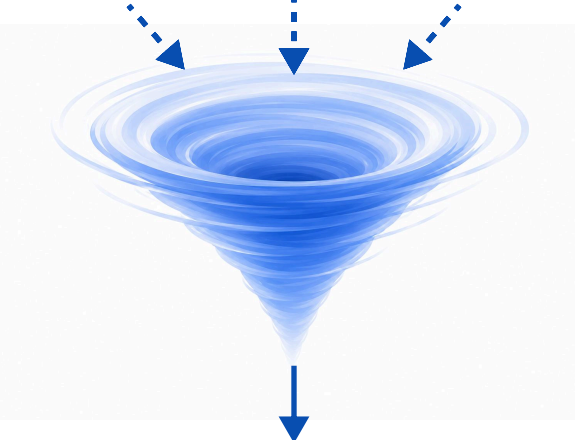
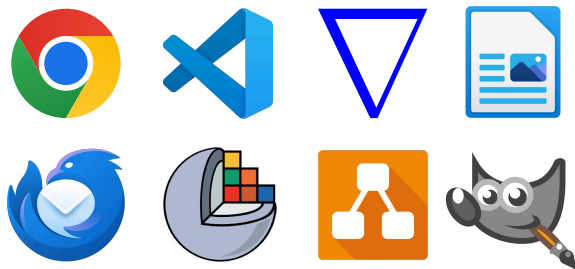
Multimedia & Design

Image Editing, Video & Audio Editing, Video Understanding, Diagram & Drawing, Format Conversion

2. Task Instances



3. GUI Use → Terminal Use



Unified Terminal Environment

```
$ ls
$ input.txt      data.csv
$ reference.png  script.sh
$ git status
$ ...
```

4. Execute in Terminal

```
$ ./run_tua_bench.sh
$ 86/120          00:53:17
agent running...
```

5. Task Verification



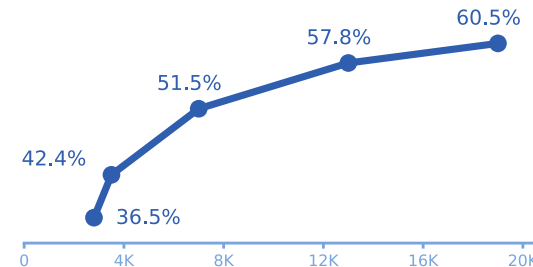
Check correctness against ground truth

- ✓ Output file existence
- ✓ Format validation
- ✓ Value accuracy
- ✓ ...

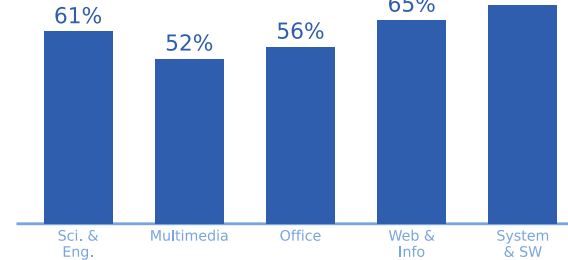
Success Rate: **53.6%**

6. Results & Analysis

Thinking-effort Scaling



Success Rate by Task Category



Cost Efficiency

